Weijia Zhang

+1 2172003915 | zhangwj.charlie@gmail.com | linkedin.com/in/weijia-charlie-zhang/ | charliedreemur.github.io | Google Scholar

Education

Yale University Sep 2026 - May 2028
M.S. in Computer Science (Thesis Track with Full Scholarship) *GPA:4.0/4.0*

University of Illinois Urbana-Champaign (UIUC) Aug 2022 - May 2026
B.S in Computer Science and Mathematics (C.W. Gear Outstanding Undergraduate Award, 1 of 2/yr) *GPA:3.7/4.0 (2025 Dean's List)*
UIUC MechMania AI Hackathon — Rank 1st

Work Experience

TikTok Jun 2026 – Sep 2026
Machine Learning Engineer Intern, RSI/Self-Evolving Agents

- As project lead, designed and shipped TikTok's agent self-evolution framework from 0 to 1 as a **shared infrastructure serving multiple production pipelines**, built on a closed *Solver–Reflector–Evolver* loop: the Solver executes tasks, the Reflector attributes failures, and the Evolver iterates reusable skills; deployed to date on unoriginal-video detection (9,600 real short videos, F1 from **62.7% to 81.3%**) and reposting-account detection (200 accounts, F1 from **76.5% to 87.3%**)
- Trained a **video-understanding VLM**: built an account-level SFT dataset of 74K accounts and 1.33M videos, and ran distributed full-parameter SFT of Qwen3-VL 2B/4B/8B on 32 H100s with DeepSpeed, plus cross-domain evaluation and threshold calibration over 53 checkpoints; the final 4B model cut false positives by **42.8%** (318 → 182) against the 8B baseline at ∼70% recall

Microsoft Jul 2025 – Sep 2025
Research Intern, Large Language Models

- Built an **execution-based RL environment** end to end to teach Excel Copilot's coding agent **newly added NLP APIs**: turned **20K+ real-world workbooks** into verifiable tasks, ran agent-written Office.js on **sandboxed workbooks** in a multi-turn **write–execute–repair** loop, and scored episodes with **final sheet-state assertions** plus **reward-hack checks**
- Post-trained the agent in this environment with **rejection-sampling SFT followed by RL** on the execution-verified reward, keeping tasks with mixed pass/fail rollouts; improved held-out **pass@1 by 15%** over the production baseline

Reborn Network May 2023 – Jul 2023
AI Agent Engineer

- Built an **embodied role-playing agent** in a Unity VR environment, closing a real-time perception–dialogue–action loop across text, voice, and full-body VR actions at sub-second end-to-end latency
- Introduced RAG/vector database and dual-level (episodic + semantic) memory, improving **cross-session recall accuracy from 38% to 61%** on an internal multi-session dialogue eval

Publications (200+ Citations)

Weijia Zhang et al. "CUADebug: Diagnosing and Repairing Computer-Use Agent Failures" *EMNLP 2026, findings.*

Weijia Zhang et al. "Is LLM's Moral Decision Compositional?" *arXiv:2606.11232. Under review at NAACL 2027.*

Weijia Zhang* and Kangcheng Luo* et al. "SkillACE: Fine-Grained RSI Agent Skill Evolution via an Adaptive Error Taxonomy." *Under review at NAACL 2027.*

Weijia Zhang* and Gordon Dai* et al. "Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory." *AI Behavioral Science Workshop, ACM SIGKDD 2024.*

Chang Wang*, and **Weijia Zhang*** et al. "COGENT: Multiagent Communicate with self-coined terms" *Under review at NAACL 2027.*

Miaomiao Li et al. (including **Weijia Zhang**). "Understanding and Mitigating the Bias Inheritance in LLM-based Data Augmentation on Downstream Tasks." *ACL 2026.* **Oral.**

Kunlun Zhu et al. (including **Weijia Zhang**). "Where LLM Agents Fail and How They Can Learn From Failures." *arXiv:2509.25370. Failure Modes in Agentic AI, ICML 2026*

Kunlun Zhu et al. (including **Weijia Zhang**). "AgentDebugX: An Open-Source Toolkit for Failure Observability, Attribution, and Recovery in LLM Agents" *arXiv:2607.18754. EMNLP 2026, System Demonstration Track*

Project Experience

OpenManus (58k Stars) & OpenManus-RL (4.1k Stars) | Core Author | GitHub Jan 2025 – Jul 2025

- Core contributor to OpenManus-RL, a **verl-based Agentic RL post-training framework**, supporting SFT and online RL with **GRPO/PPO**
- Built a multi-turn agent-environment rollout pipeline that unifies **ReAct reasoning, tool actions, environment interaction, trajectory collection, and reward computation** for long-horizon tasks including GAIA, WebShop, and ALFWorld
- Curated and released **∼50K high-quality ReAct agent trajectories** spanning OS, Web, Database, E-commerce environments

CUAAgentDebugger | First Author & Lead Jan 2026 – Aug 2026

- Introduced **CUAErrorBench**, comprising **204 human-annotated failed OSWorld trajectories**, and a hierarchical CUA failure taxonomy with **5 categories and 30 subtypes** spanning perception, grounding/interaction, reasoning/control, system failures, and task infeasibility
- Built **CUADebugger**, a ReAct-style root-cause analysis agent that localizes the **earliest causal error** from trajectory screenshots and action traces, produces grounded diagnoses and corrections, and retrieves prior debugging memories to guide re-execution; improved failure recovery from **13.89% to 29.86%** and overall task accuracy from **61.77% to 68.14%**

ReGameBench: Long-Horizon Visual Coding via Game Reconstruction | First Author & Lead Aug 2026 – Present

- Leading **ReGameBench**, a benchmark for **long-horizon visual coding and game reconstruction**: 100 games with **executable game environments** and art assets, plus **2,000 gameplay videos** with paired input trajectories
- Designed an automated evaluation pipeline combining **multi-agent rubric extraction, timestamp-grounded visual requirements, held-out interaction replay, and rubric-guided VLM judging** to measure behavioral and visual fidelity

Skills

Languages: Python, C/C++, C#, TypeScript/JavaScript, SQL
ML Systems: PyTorch, DeepSpeed, verl, vLLM, Distributed Training, Docker, Linux
ML/LLM: Agentic RL, PPO/GRPO, SFT, RL Environments, Tool-Use Agents, Multimodal Agents, ReAct